

RW4 Promo-State and Discount Models

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

incidence_rows=1

type_rows=1

depth_rows=1

asymmetry_rows=422

Table: Promo_State_Incidence

standardized_type	model_scope	model_unreliable_flag	reason
ALL	unavailable	binomial_attempt_failed	1 Promo-incidence binomial ...

Table: Promo_State_Type

standardized_type	model_scope	model_unreliable_flag	reason
ALL	unavailable	multinomial_attempt_failed	1 Promo-state multinomial m...

Table: Promo_State_Depth

standardized_type	model_scope	model_unreliable_flag	reason
ALL	unavailable	depth_attempt_failed	1 Promo-depth conditional m...

Table: Asymmetry_Observed_vs_Baseline

segment_name	standardized_type	farm_gate_source	reconstruction_variant	chain_direction	stage_from	stage_to	model_family	observed_sr_coef	baseline_sr_coef	delta_sr_coef	observed_lr_coef	baseline_lr_coef	delta_lr_coef	observed_ect_coef	baseline_ect_coef	delta_ect_coef	same_sign_lr	same_sign_ect	pseudo_asymmetry_likely
Вершки	cream	initial	linear	forward	ProZorro	Retail	NARDL	0.1855195387021934	0.1007392494158206	-0.0847802892863728	0.2943590807532827	0.146246553131392	-0.1481125276218907	-0.7295217442383444	-0.7945546025645894	-0.06503285832624506	True	True	0
	cream	initial	linear	forward	FarmGateUA	ProZorro	ARDL	27.40581164655441	27.40581164655441	0.0	-1.563367219701413	-1.563367219701413	0.0	nan	nan	nan	True	False	0
	cream	initial	linear	forward	FarmGateUA	ProZorro	NARDL	-39.44395541420697	-39.44395541420697	0.0	-18.22659908428248	-18.22659908428248	0.0	-1.394427676720398	-1.394427676720398	0.0	True	True	0
	cream	initial	linear	forward	FarmGateUA	Retail	NARDL	-8.772023103981054	-4.075323960016448	4.696699143964606	-1.852043589957451	-2.798583866922371	-0.94654027696492	-0.7869901901005358	-0.9482306245503481	-0.1612404344498123	True	True	1
	cream	initial	linear	reverse	Retail	ProZorro	NARDL	1.562917141304475	2.310060540554998	0.7471433992505232	-0.2420696049936481	0.694874012870825	0.936943617864473	-1.11930546143876	-1.131840757655833	-0.012535296217072833	False	True	1
Вершки	cream	initial	linear	reverse	ProZorro	FarmGateUA	NARDL	0.001294370750072191	0.001294370750072191	0.0	0.02102571537973896	0.02102571537973896	0.0	-0.1349163065634996	-0.1349163065634996	0.0	True	True	0
Вершки	cream	initial	linear	reverse	Retail	FarmGateUA	NARDL	0.006744790717442998	-0.01054007989158433	-0.01728487060902733	0.0979596312980813	0.1881569946873936	0.09019736338931231	-0.144872313268366	-0.1099961288446253	0.0348761844237407	True	True	0
Молоко питне	milk	initial	linear	forward	ProducerUA	ProZorro	NARDL	-39.35627563072685	-39.35627563072685	0.0	-1.428284989733576	-1.428284989733576	0.0	-0.9847510668431191	-0.9847510668431191	0.0	True	True	0
Молоко питне	milk	initial	linear	forward	ProZorro	Retail	NARDL	-0.05150967266593062	-0.07759645603786203	-0.026086783371931405	-0.04955189374108957	-0.1875189372472206	-0.13796704350613104	-0.5842457751880188	-0.5984997219947372	-0.014253946806718432	True	True	0
Молоко питне	milk	initial	linear	forward	ProducerUA	Retail	NARDL	-15.29852457315202	16.08250489949674	31.381029472648763	1.143785547528498	5.19149651385324	4.0477109663247415	-0.7215120205461834	-0.9142138640633395	-0.19270184351715614	True	True	1
Молоко питне	milk	initial	linear	reverse	Retail	ProZorro	NARDL	-3.377883072190306	-4.31932444960139	-0.9414414238235942	-0.8065514462952509	-1.65425420802465	-0.8477027617293992	-0.9583680273094293	-1.048219149760883	-0.08985112245145377	True	True	1
Молоко питне	milk	initial	linear	reverse	ProZorro	ProducerUA	ARDL	-0.003981249234879858	-0.003981249234879858	0.0	0.08556001807837454	0.08556001807837454	0.0	nan	nan	nan	True	False	0
Молоко питне	milk	initial	linear	reverse	ProZorro	ProducerUA	NARDL	-0.06869734434387867	-0.06869734434387867	0.0	-0.7347687591369463	-0.7347687591369463	0.0	-0.1809708253238888	-0.1809708253238888	0.0	True	True	0
Молоко питне	milk	initial	linear	reverse	Retail	ProducerUA	NARDL	-0.2289575426204663	0.03043734523875096	0.25939488785921727	-4.567563531007199	4.632234498859097	9.199798029866296	-0.05896759892425604	-0.04125585800661566	0.017711740917640384	False	True	1
Сир твердий	hard_cheese	initial	linear	forward	FarmGateUA	ProducerUA	NARDL	0.9718885163453532	0.9718885163453532	0.0	4.164046497719572	4.164046497719572	0.0	-0.02099956516295825	-0.02099956516295825	0.0	True	True	0
Сир твердий	hard_cheese	initial	linear	forward	ProducerUA	ProZorro	ARDL	2.533791668492441	2.533791668492441	0.0	0.8515901678330565	0.8515901678330565	0.0	nan	nan	nan	True	False	0
Сир твердий	hard_cheese	initial	linear	forward	ProducerUA	ProZorro	NARDL	-81.10378249455037	-81.10378249455037	0.0	-3.429536449371221	-3.429536449371221	0.0	-0.913365423696856	-0.913365423696856	0.0	True	True	0
Сир твердий	hard_cheese	initial	linear	forward	ProZorro	Retail	ARDL	-0.008082951427838936	-0.2316252020709719	-0.22354225064313296	0.1583981708763298	0.08260811044262882	-0.07579006043370097	nan	nan	nan	True	False	1
Сир твердий	hard_cheese	initial	linear	forward	ProZorro	Retail	NARDL	0.01249324517355312	-0.3638721758900579	-0.376365421063611	0.2033362629705365	-0.4370611614962356	-0.6403974244667721	-1.086095722865538	-0.9455579020345951	0.1405378208309428	False	True	1
Сир твердий	hard_cheese	initial	linear	forward	FarmGateUA	ProZorro	NARDL	-68.77473824351345	-68.77473824351345	0.0	-11.37819556032246	-11.37819556032246	0.0	-1.331056770131037	-1.331056770131037	0.0	True	True	0
Сир твердий	hard_cheese	initial	linear	forward	ProducerUA	Retail	NARDL	17.92245839749829	51.00085056474553	33.07839216724724	5.795116967333372	8.708316733703773	2.9131997663704015	-0.9697285021614469	-1.115149434966051	-0.145420932804604	True	True	1
Сир твердий	hard_cheese	initial	linear	forward	FarmGateUA	Retail	ARDL	-16.12941410918768	14.40821465960636	30.537628768794043	-3.273094954784757	-1.000896839545215	2.272198115239542	nan	nan	nan	True	False	1
Сир твердий	hard_cheese	initial	linear	forward	FarmGateUA	Retail	NARDL	-15.9250044387613	4.298851419629319	20.22385585839062	-6.898921926023638	-7.253522207662531	-0.3546002816388931	-1.261934327941142	-1.202840410988928	0.059093916952214	True	True	1
Сир твердий	hard_cheese	initial	linear	reverse	Retail	ProZorro	NARDL	-0.5907839473896122	-1.121124059021263	-0.5303401116316508	-0.2203690553083897	-1.338523197029263	-1.1181541417208734	-0.6560675361469568	-0.8410859443829597	-0.1850184082360029	True	True	1
Сир твердий	hard_cheese	initial	linear	reverse	ProZorro	ProducerUA	ARDL	0.05875093381259333	0.05875093381259333	0.0	-0.0758952357905691	-0.0758952357905691	0.0	nan	nan	nan	True	False	0
Сир твердий	hard_cheese	initial	linear	reverse	ProZorro	ProducerUA	NARDL	0.02086721830064114	0.02086721830064114	0.0	0.5045181895255855	0.5045181895255855	0.0	-0.06470054551241512	-0.06470054551241512	0.0	True	True	0
Сир твердий	hard_cheese	initial	linear	reverse	Retail	ProducerUA	NARDL	0.02603038881447416	0.1166401499278574	0.09060976111338324	-0.122865301952281	-2.011505148704162	-1.888639846751881	-0.04487768835028339	-0.02830467340125044	0.01657301494903295	True	True	0
Сир твердий	hard_cheese	initial	linear	reverse	ProducerUA	FarmGateUA	NARDL	1.220088115787272	1.220088115787272	0.0	-0.02224966033766555	-0.02224966033766555	0.0	-0.07024274699822596	-0.07024274699822596	0.0	True	True	0
Сир твердий	hard_cheese	initial	linear	reverse	ProZorro	FarmGateUA	NARDL	-0.009622956572284569	-0.009622956572284569	0.0	-0.1282890618100976	-0.1282890618100976	0.0	-0.1551583039762704	-0.1551583039762704	0.0	True	True	0
Сир твердий	hard_cheese	initial	linear	reverse	Retail	FarmGateUA	ARDL	-0.004884939285385581	0.01173175474368326	0.01661669402906884	-0.08156888394341669	0.5752798070776506	0.6568486910210672	nan	nan	nan	False	False	1
Сир твердий	hard_cheese	initial	linear	reverse	Retail	FarmGateUA	NARDL	0.001364224083898027	0.01421478072341609	0.012850556639518064	-0.006678367612549292	0.2315380381811967	0.23821640579374598	-0.1385585915307108	-0.09117603112682349	0.0473825604038873	False	True	1
Сметана	sour_cream	initial	linear	forward	ProZorro	Retail	NARDL	-0.01629282865575449	-0.06636167401275544	-0.05006884535700096	0.2251736657556283	0.02591465193697578	-0.19925901381865252	-0.7929811801699767	-0.7337665367972408	0.05921464337273585	True	True	0
Сметана	sour_cream	initial	linear	forward	FarmGateUA	ProZorro	NARDL	-33.39030946236475	-33.39030946236475	0.0	-4.64565248280245	-4.64565248280245	0.0	-0.9480197072995824	-0.9480197072995824	0.0	True	True	0
Сметана	sour_cream	initial	linear	forward	FarmGateUA	Retail	NARDL	-13.71924367257179	-7.141058680813738	6.578184991758053	-3.081786729405028	-5.89671224769782	-2.8149255182927924	-0.7511669760850768	-0.839933184232939	-0.08876620814786218	True	True	1
Сметана	sour_cream	initial	linear	reverse	Retail	ProZorro	NARDL	-0.3814286047977202	-0.4868810925947405	-0.10545248779702027	0.09087453878412201	-0.7561698943478722	-0.8470444331319942	-0.7897667964593056	-0.7861731847468327	0.003593611712472855	False	True	1
Сметана	sour_cream	initial	linear	reverse	ProZorro	FarmGateUA	ARDL	-0.0002561523368839191	-0.0002561523368839191	0.0	-0.007181451310785285	-0.007181451310785285	0.0	nan	nan	nan	True	False	0
Сметана	sour_cream	initial	linear	reverse	ProZorro	FarmGateUA	NARDL	0.002734014348778207	0.002734014348778207	0.0	-0.00205895003852816	-0.00205895003852816	0.0	-0.1823815878824159	-0.1823815878824159	0.0	True	True	0
Сметана	sour_cream	initial	linear	reverse	Retail	FarmGateUA	NARDL	0.002067069993916146	0.006396298825528593	0.004329228831612447	0.03418390263543871	0.05454611122925054	0.020362208593811833	-0.1563619672392225	-0.1420033484411655	0.014358618798056999	True	True	0
all_products_average	all_products_average	initial	linear	forward	FarmGateUA	ProducerUA	NARDL	0.04224957586189704	0.04224957586189704	0.0	-4.851556665263181	-4.851556665263181	0.0	-0.01119891013625641	-0.01119891013625641	0.0	True	True	0
all_products_average	all_products_average	initial	linear	forward	ProducerUA	ProZorro	NARDL	144.5589989058615	144.5589989058615	0.0	-4.789181955386666	-4.789181955386666	0.0	-1.20953796040727	-1.20953796040727	0.0	True	True	0

Table: Discount_Strategy_Synthesis

question	answer	evidence
Do promotions behave like...	yes	Baseline Silpo ECM-style ...
Does pseudo-asymmetry wea...	mixed	Observed vs baseline coef...
Do effective prices adjus...	mixed	share_observed_ect_faster...
Does discount depth react...	mixed	Promo_State_Depth stores ...
Do retail shocks transmit...	yes	reverse_core_rows=32
Is the retail-to-farmgate...	yes	reverse_robust_share=1.00
Is transmission stronger ...	yes	product_core_rows=94; ave...
Do specific retailer bran...	yes	brand_core_rows=55

Observed vs Baseline Short-Run Delta

